

# Project Closure Report

## Advanced Chooser Option Pricing Model

**Underlying** JPMorgan Chase and Co  
**Project period** Week 1 through Week 8  
**Closure date** 2026-09-08

### Closure Summary

The eight-week project produced an end-to-end chooser-option research pipeline that connects market data collection, preprocessing, Black-Scholes-Merton pricing, model validation, machine learning, sensitivity analysis, and an interactive Streamlit prototype. The technical work is complete and archived. Final public deployment and the narrated demonstration remain student-controlled submission actions.

The selected production route forecasts volatility with Random Forest and then applies the validated chooser BSM formula. On the held-out proxy target, this route reduced RMSE by 11.30% relative to the trailing-volatility BSM baseline. The direct-price model was retained as a documented extrapolation failure.

### Project Objectives and Outcomes

| No | Objective                                  | Closure outcome                                                                         |
|----|--------------------------------------------|-----------------------------------------------------------------------------------------|
| 1  | Replicate and validate a BSM chooser model | Completed with paper-matched parameters and numerical checks                            |
| 2  | Build a multidimensional market dataset    | Completed using JPM, VIX, and Treasury data from 2018 to 2024                           |
| 3  | Develop machine-learning extensions        | Completed through volatility forecasting and direct pricing routes                      |
| 4  | Compare ML with the BSM baseline           | Completed on one chronological held-out test period                                     |
| 5  | Package a usable pricing tool              | Completed locally and pushed to a private GitHub repository; public app hosting remains |

## Eight Week Delivery Record

| Period | Archived work                                                                    | Status                                                                                 |
|--------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Week 1 | Data specification, API notebook, raw JPM, VIX, and Treasury data                | Complete                                                                               |
| Week 2 | Cleaned 23-column dataset, automated preprocessing, engineered features          | Complete; scheduled GitHub Actions passed; news-sentiment gap recorded                 |
| Week 3 | Chooser BSM notebook, validation notebook, parameter configuration               | Complete                                                                               |
| Week 4 | Baseline evaluation, validation report, performance benchmark, reusable BSM code | Complete with transparent proxy and listed-option validation                           |
| Week 5 | ML architecture, optimized features, two initial modeling approaches             | Complete                                                                               |
| Week 6 | Trained .pkl and .joblib models, performance comparison, SHAP analysis           | Complete                                                                               |
| Week 7 | Sensitivity and stress analysis, pricing prototype, data updater                 | Complete with documented sentiment-data limitation                                     |
| Week 8 | Packaged pricing tool, final technical report, presentation deck                 | Technical package and private GitHub repository complete; public URL and video pending |

## Data and Model Chain

- Week 1 raw market data flows into the Week 2 preprocessing and feature pipeline.
- Week 3 supplies the validated chooser BSM implementation and contract configuration.
- Week 4 evaluates that model historically and under listed-option sanity checks.
- Weeks 5 and 6 train, select, compare, and serialize the ML extensions.
- Weeks 7 and 8 add stress testing, data updates, presentation materials, and the user interface.

## Key Technical Results

| Evidence                 | Result                       | Interpretation                                  |
|--------------------------|------------------------------|-------------------------------------------------|
| Processed market dataset | 1,760 rows and 23 columns    | Aligned JPM, VIX, and Treasury history          |
| Paper choice decisions   | 10 of 10 matched             | Week 3 replication check                        |
| Put-call parity error    | 0.000000                     | Analytical consistency check                    |
| Monte Carlo comparison   | 0.823 standard errors        | Closed-form price within simulation uncertainty |
| Selected ML route        | RMSE 4.3203 and R2 0.9575    | Random Forest volatility plus BSM               |
| BSM baseline             | RMSE 4.8706 and R2 0.9460    | Same 252 held-out dates                         |
| Direct-price model       | RMSE 49.9891 and R2 - 4.6915 | Rejected for extrapolation failure              |

## Model Delivery and Reproducibility

Week 6 now includes native Python pickle files for both selected estimators, as required by the project outline. Each file was reloaded and checked on all 252 held-out rows. Joblib copies are retained for the packaged Week 8 application, and model metadata records the feature order, contract assumptions, evaluation dates, artifact manifest, and round-trip validation results.

## Stress Test Record

| Scenario                       | Chooser price | Change     |
|--------------------------------|---------------|------------|
| Base                           | \$99.2617     | \$0.0000   |
| 50 percent volatility spike    | \$115.6683    | +\$16.4066 |
| Two percentage point rate hike | \$101.3317    | +\$2.0700  |
| Combined stress                | \$116.9177    | +\$17.6561 |

## Limitations and Closure Controls

| Limitation                                                   | Control                                                                   |
|--------------------------------------------------------------|---------------------------------------------------------------------------|
| No public historical chooser transaction series              | All chooser accuracy scores are labeled as theoretical proxy results      |
| CME wording does not align cleanly with a JPM equity chooser | Cboe listed JPM vanilla quotes are used only as a BSM market sanity check |
| No reproducible news sentiment series                        | The gap is disclosed; VIX is not relabeled as sentiment                   |
| Dividend yield was not collected historically                | The paper value of 2.33 percent remains an explicit assumption            |
| Tree models have weak extrapolation                          | BSM remains visible and the interface warns outside fitted ranges         |
| Serialized estimators require a trusted environment          | Load only project-owned files and preserve documented package versions    |

## Archive Location

All formal reports are consolidated under the project Reports directory, organized by weekly subfolders. This closure report is stored at the Reports root for direct archival access. Source code, datasets, notebooks, model artifacts, figures, and the deployable application remain in their week-specific project directories.

## GitHub Delivery

The complete project is committed to <https://github.com/Frank-Wang-Jc/jpm-chooser-option-pricing>. Automated checks passed for repository integrity, application startup, and the scheduled data preprocessing pipeline.

## Outstanding Submission Actions

- Deploy the Streamlit application, record the public URL, and repeat the smoke test in a private browser window.
- Record the required narrated 5 to 10 minute demonstration video using the prepared demo script.
- Before submission, make the private repository public or grant the reviewer access. Credentials, local environments, licensed references, and duplicate archives remain excluded.

## Closure Statement

The project has reached technical completion across the planned eight-week sequence. Its code, data, validation evidence, trained models, reports, presentation, and local application are organized for review and archival. Closure becomes administratively complete after the public deployment links and narrated demonstration are added to the submission package.